

Powers of Products and Quotients

Simplify. Your answer should contain only positive exponents.

1) $(3a^2)^3$

$27a^6$

2) $(2n^4)^4$

$16n^{16}$

3) $(3x^4)^4$

$81x^{16}$

4) $(6b^2)^2$

$36b^4$

5) $(7y^4)^2$

$49y^8$

6) $(3ab^4)^4$

$81a^4b^{16}$

7) $(2x^4y^4)^3$

$8x^{12}y^{12}$

8) $(5mn^3)^3$

$125m^3n^9$

9) $(x^2y^2)^2$

x^4y^4

10) $(6yx^4)^2$

$36y^2x^8$

11) $(u^4v^3)^2$

u^8v^6

12) $(2x^4y^4)^4$

$16x^{16}y^{16}$

13) $(3x^2 \cdot 2x^2)^2$

$36x^8$

14) $(2p^3 \cdot 2p)^2$

$16p^8$

15) $(4n^3 \cdot n^2)^2$

$16n^{10}$

16) $(3x \cdot 2x)^2$

$36x^4$

17) $(4x^4 \cdot x^4)^3$

$64x^{24}$

18) $(4n^4 \cdot n)^2$

$16n^{10}$